

Software Test Plan (STP)

Hostel Management System (HMS)

Project: Hostel Management System

Version: 1.0

Authors: <QA Team>

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Status: Sample

1. Introduction

Purpose: This Software Test Plan defines the scope, strategy, environment, resources, schedule approach, deliverables, risks and acceptance criteria for testing HMS Release 1.0. Tests shall be traceable to the unique requirement identifiers in the HMS SRS, including functional requirements (FR), non-functional requirements (NFR), security requirements (SEC-REQ), business rules (BR) and user-interface requirements (UI).

Scope: HMS is a responsive web application used by Students, Wardens and Administrators to manage student profiles, hostel/room inventory, room applications and bed allocation, room changes and check-out, complaints, fee records, reports, notifications and announcements.

- In scope: all functional requirements FR-1.1 through FR-9.3; interface requirements UI-1 through UI-8; business rules BR-1 through BR-6; performance, safety, security, usability, accessibility, availability, reliability, maintainability, portability and scalability requirements stated in SRS Sections 5.1–5.5.
- Out of scope: online payment collection or payment-gateway processing, mess/menu management, visitor and leave/out-pass management, biometric attendance, a native mobile application, physical hostel maintenance and third-party service internals. HMS records payments made through existing university channels; it does not collect them.
- External interfaces such as the university SMTP relay and CSV student master-list import will be tested at the HMS boundary, using controlled test data and test/stub services where required.

References: Hostel Management System SRS v1.0, Design Specifications v1.0, Database Design Document and Software Requirements Specification – Hostel Management System, Version 1.0 approved, PES University, Department of CSE, 04 October 2026.

Definitions:

Term	Meaning
HMS	Hostel Management System
SRS	Software Requirements Specification
STP	Software Test Plan
FR / NFR	Functional Requirement / Non-functional Requirement
SEC-REQ	Security Requirement
BR	Business Rule
UI	User Interface requirement
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
SRN	Student Registration Number
RBAC	Role-Based Access Control
IDOR	Insecure Direct Object Reference / object-level access control failure

2. Test Items

- Authentication and access control: registration, email verification, login, password reset, lock-out, logout, session expiry and role permissions.
- Student profile management and student search, including field validation, photo upload, hostel-scoped search and audit logging.
- Hostel and room management: hostel/room CRUD with soft-deactivation, CSV room import, room capacity/status and availability grid.
- Room application and allocation: application window, one application per academic year, preferences, application queue, allocation priority, manual/automatic allocation and concurrency protection against double allocation.
- Room changes requests, approval/rejection, check-out, allocation history and bed release.
- Complaints: creation, ID generation, permitted status transitions, remarks, reopening window and overdue dashboard alerts.
- Fee tracking: fee generation, payment validation, balances, allocation confirmation, PDF receipts and overdue provisional allocations.
- Role-specific dashboards, reports, filters, CSV/PDF export and emergency resident list.

- Notifications and announcements: event emails, retry/logging behavior and announcement expiry.
- Cross-cutting requirements: HTTPS/TLS, session and request security, audit logs, upload validation, data minimization, backups/restore, performance, usability, accessibility, browser compatibility and responsive layouts.

3. Features to be Tested

Features mapped to SRS requirement IDs:

- HMS-F-001: Authentication and access control
- HMS-F-002: Student profile management and search
- HMS-F-003: Hostel and room management
- HMS-F-004: Room applications and allocation
- HMS-F-005: Room changes and student check-out
- HMS-F-006: Complaint management and tracking
- HMS-F-007: Fee tracking, payments and receipts
- HMS-F-008: Dashboards and report generation
- HMS-F-009: Email notifications and announcements
- HMS-F-010: User interface and form validation
- HMS-NF-001: Performance and response time
- HMS-NF-002: Security and access control
- HMS-NF-003: Backup, recovery and data reliability
- HMS-NF-004: Usability and accessibility
- HMS-NF-005: Availability and maintainability
- HMS-NF-006: Browser compatibility and responsiveness
- HMS-NF-007: Scalability and concurrent-user support

4. Features Not to be Tested

- Actual collection or settlement of fees through online payment gateways; only recording and validating payment records in HMS is tested.
- Mess/menu management, visitors, leave/out-pass workflows, biometric attendance and native mobile app functionality, because these are explicitly out of scope for Release 1.0.
- Internal operation of the university student-records system and SMTP provider; only HMS import/send behavior and error handling at the interface are tested.
- Physical hostel conditions, room inspections, client hardware failures and general network infrastructure beyond checks needed to validate HMS HTTPS connectivity.

5. Test Approach / Strategy

Levels:

- Unit testing (individual modules such as registration and room allocation)
- Integration testing (student module, room module, and database)
- System testing (end-to-end hostel management functionality)
- Acceptance testing (UAT by hostel administrators and selected students)

Types:

- Functional and negative testing: valid/invalid inputs, boundary values, permissions, duplicate actions and expected error messages.
- Regression testing: rerun affected test cases after fixes or changes to authentication, allocation, fee, complaints, reports or permissions.
- Database integrity and concurrency testing: capacity/status consistency, one active allocation per student/bed, atomic room changes and simultaneous allocation attempts.
- Performance/load testing: use Locust as specified by the SRS; verify all NFR-PERF thresholds under the defined data volume and concurrent-user load.
- Security testing: role/hostel-scope enforcement, IDOR checks, password policy/hash storage, lock-out/rate limits, session cookies, HTTPS/HSTS, CSRF, XSS/SQL injection protection, upload validation, audit-log permissions and secrets handling.
- Usability/accessibility/compatibility testing: first-time application task, WCAG 2.1 AA checks, keyboard navigation, screen reader/text alternatives, contrast, current two releases of supported browsers and 360–1920 px widths.
- Backup/recovery and reliability testing: daily backup schedule/retention, restore drill, consistency checks and soft-delete recovery.

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Verify password hashing and secure password storage.
- Test role-based access control for students and administrators.
- Validate input fields against SQL injection and cross-site scripting (XSS).
- Ensure student personal information is accessible only to authorized users.

- Verify session timeout and logout functionality.
- Verify server-side authorization on every request according to BR-1; unauthorized role or hostel-scope access returns HTTP 403 or 404 as specified and is logged. Change IDs in URLs to test for IDOR.
- Verify uploads by file content/magic bytes and size (JPEG/PNG/PDF, maximum 2 MB), storage outside web root, random file names and authorized delivery; verify student data export and data minimization.

Exit / Acceptance Criteria

All planned high-priority and release-critical test cases have been executed; execution, pass/fail, blocked and not-run status is recorded against each test case.

No open Critical or High severity defect remains for release sign-off; lower-severity defects have documented disposition and approval.

All mandatory SRS acceptance thresholds are met or deviations are formally documented and approved. In particular, allocation correctness and access control must not be waived.

The OWASP ZAP baseline scan has zero High/Critical findings; performance, backup/restore, usability and compatibility results are documented.

Requirements Traceability Matrix is updated with test results and evidence; UAT sign-off is recorded by the designated project/customer representatives.

6. Test Environment

Area	Specification / Setup
Server	Ubuntu Server 24.04 LTS; Python 3.12; Django 5.x served by Gunicorn behind Nginx; reference configuration 4 vCPU, 8 GB RAM and 100 GB SSD.
Database	PostgreSQL 16; psycopg 3; database access restricted to the server/private network.
Client/browser	Latest two releases of Google Chrome, Mozilla Firefox, Microsoft Edge and Apple Safari; widths from 360 px to 1920 px.
Network/security	HTTPS/TLS 1.2 or higher; HTTP port 80 redirects to HTTPS; test TLS/HSTS and SMTP STARTTLS configuration.

Interfaces	Test SMTP relay or controlled mail capture; CSV import fixtures; PDF receipt/report generation via ReportLab; Bootstrap 5.3 UI.
Tools	Locust for load testing; OWASP ZAP baseline scan; browser developer tools; automated unit-test and coverage tools; flake8; spreadsheet/CSV and PDF validation tools; defect tracker selected by the team.
Test data	Synthetic student records with SRN, name, gender, programme, year and institutional email; student profiles; role-specific accounts; hostels, room types, capacity and fees; applications with priority categories; complaints; payment histories and announcements.

7. Test Schedule

Milestones:

- Test case design: 05-Oct-2026
- Environment setup: 07-Oct-2026
- Test execution start: 08-Oct-2026
- Test execution end: 20-Oct-2026
- UAT: 22-Oct-2026 to 25-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (RTM)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Prema P Kotur	Prepare plan, coordinate execution

Test Engineer	Ojas Taori	Design & execute test cases, log defects
Developer	Praakruthi PS	Support defect fixes and triage
Product Owner	Raina Kareem	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Impact	Mitigation
Unstable or late build	Testing delayed or invalidated	Request incremental builds; run smoke tests before full execution.
Incomplete or ambiguous requirement interpretation	Inconsistent expected results	Use SRS IDs and business rules; record clarification decisions before finalizing cases.
Incorrect allocation or race conditions	Double booking or incorrect hostel assignment	Test concurrent requests, database constraints, transactions, gender matching and consistency checks.
Sensitive test data exposure	Privacy/security incident	Use synthetic data, restrict access and verify logs/files do not expose sensitive values.
SMTP service unavailable	Email scenarios blocked	Use a test relay/mail capture; separately test retry and final-failure logging.
Load/security tools unavailable or misconfigured	Non-functional requirements not verified	Validate tools and environment early; record any unexecuted requirement as a release limitation.
Backup cannot be restored	Potential data loss	Test restoration monthly as required; verify 30-day retention and off-machine storage.
Cross-browser or mobile-width layout defect	Student workflow inaccessible	Test all specified browser families and widths, including 360 px.

11. Assumptions & Dependencies

- The university provides a student master list CSV before each application window, including the fields defined by the SRS.
- Test accounts use working institutional email addresses, and a university SMTP relay or approved test alternative is available.
- The IT department supplies the server VM, domain and TLS certificate; the deployment matches the SRS environment closely enough for meaningful performance/security results.
- Hostel rules, room fees, gender designations, priority categories and application-window dates are supplied by the hostel office.
- Fee collection occurs outside HMS; test cases verify recording, validation, balances and receipts only.
- The implementation uses Django 5.x, PostgreSQL 16, server-rendered Django templates and Bootstrap 5 as specified in the SRS.

12. Suspension & Resumption Criteria

12.1 Suspend testing when

- The test environment or database is unavailable or data corruption makes results unreliable.
- A blocking defect prevents execution of a major workflow such as authentication, room application/allocation, fee recording or role authorization.
- The build is unstable enough that a substantial portion of planned cases cannot be executed.
- Test data or external test interfaces required for a test are unavailable and no safe stub/workaround exists.

12.2 Resume testing when

- The blocking defect is fixed or an approved workaround is available and verified.
- The environment is stable, test data has been restored/reset and smoke tests pass.
- The team has received a verified build and the affected test cases/regression scope have been identified.

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) ensures that SRS requirements are mapped to their corresponding test cases.

Example:

- **HMS-F-001 (Student registration and authentication)** → TC-01, TC-02, TC-03
- **HMS-F-010 (Student details management and search)** → TC-05, TC-06
- **HMS-F-020 (Room allocation and room changes)** → TC-08, TC-09, TC-10, TC-11, TC-12
- **HMS-F-030 (Room availability and management)** → TC-07
- **HMS-F-040 (Hostel fee management and receipts)** → TC-14
- **HMS-F-050 (Dashboards, reports and exports)** → TC-15
- **HMS-F-060 (Complaint management)** → TC-13
- **HMS-F-070 (Email notifications)** → TC-16
- **HMS-NF-001 (Performance and response time)** → TC-17
- **HMS-NF-002 (Security and access control)** → TC-04, TC-19, TC-20, TC-21
- **HMS-NF-003 (Backup and recovery)** → TC-18
- **HMS-NF-004 (Audit logging)** → TC-22
- **HMS-NF-005 (Usability and accessibility)** → TC-23
- **HMS-NF-006 (Availability, maintainability and portability)** → Additional test cases to be assigned
- **HMS-NF-007 (Scalability and concurrent-user support)** → Additional test cases to be assigned

The RTM also covers remaining functional requirements, business rules, UI requirements, and security requirements to ensure complete test coverage.

14. Test Metrics & Reporting

Metrics collected:

Test cases planned, executed, passed, failed, blocked and not run; execution and pass percentages.

Requirement coverage: number and percentage of applicable SRS requirements mapped to test cases and executed.

Defect counts by severity/priority, open/closed status, defect aging and reopen rate.

Performance metrics against each NFR-PERF threshold, including response-time percentiles and throughput under load.

Security findings by severity and status; OWASP ZAP High/Critical finding count.

Unit-test statement coverage for allocation, fee and access-control modules; flake8 result.

Availability, backup/restore, accessibility, compatibility and usability evidence.

Reports:

- Daily test execution status
- Defect status reports
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Prema P Kotur	Prema P Kotur/6-10-2026
Dev Lead	Praakruthi PS/Ojas Taori	Praakruthi PS/Ojas Taori/6-10-2026
Product Owner	Raina Kareem	Raina Kareem/ 6-10-2026